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Travelling Salesperson Problem (TSP)

Exact Algorithms for TSP

Video:

Held and Karp's Dynamic Programming Algorithm

32 min

Video:

Integer Linear Programming Formulation

21 min

Video:

Subtours and Subtour Elimination Formulation

13 min

Lab:

Interactive Notes on Exact Approaches to TSP

1h

Quiz:

Held-Karp Algorithm

Submitted

Quiz:

TSP Integer Programming

30 min

Approximation Algorithm for TSP

Heuristics: A Brief Tour

Approximation Schemes and the Knapsack Problem

Problem Set # 4

Held-Karp Algorithm

Review Learning Objectives

Submit your assignment

Due

Mar 3, 11:59 PM IST

Receive grade

To Pass

80% or higher

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Grade

received 100%

Latest Submission

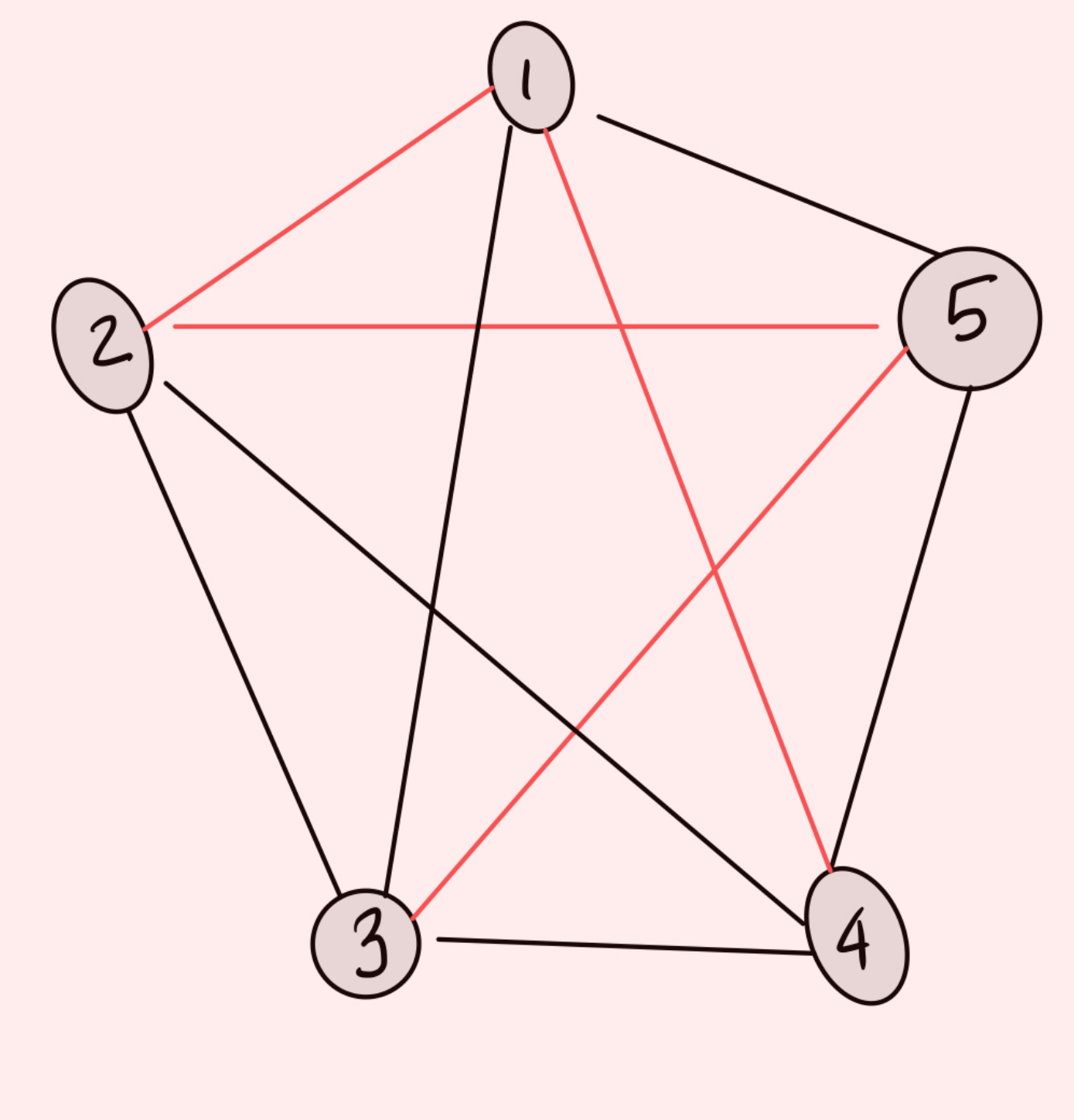
Grade 100%

To pass 80% or

higher

Go to next item

The problem below concerns the following TSP instance. Edges shown in black have a weight of 1 whereas red edges have a weight of 3.



Assume vertex 1 is the designated start/end point of the tour. Let C_{ij} represent the cost of the edge from vertex i to vertex j .

1. Which of the following expressions is the appropriate equation for the value of $\text{minCostPath}(\{2, 3\}, 4)$ 3 / 3 points

- ☒ $\min(\text{minCostPath}(\{2\}, 3) + C_{3,4}, \text{minCostPath}(\{3\}, 2) + C_{2,4})$
- ☐ $\max(\text{minCostPath}(\{2\}, 3) + C_{4,3}, \text{minCostPath}(\{3\}, 2) + C_{2,3})$
- ☐ $\min(\text{minCostPath}(\{2\}, 4) + C_{4,3}, \text{minCostPath}(\{3\}, 4) + C_{4,2})$
- ☐ $\min(\text{minCostPath}(\{2, 4\}, 3), \text{minCostPath}(\{3, 4\}, 2))$
- ☒ Correct

2. Calculate the value of $\text{minCostPath}(\{2, 3\}, 4)$. 3 / 3 points

3

- ☒ Correct
- Use previous problem to expand the recurrence correctly and then note that for a singleton set $\text{minCostPath}(\{i\}, j) = C_{1i} + C_{ij}$. Use a pen and paper and calculate carefully.

3. What vertex in the graph would you associate with the memo table corresponding to $\text{minCostPath}(\{2, 3\}, 4)$ to recover the minimum cost path? 2 / 2 points

- ☒ 3
- ☐ 2
- ☐ 4
- ☒ Correct
- Correct : The choice of vertex 3 as the vertex to be visited before vertex 4 yields the optimal solution for this case.

4. Consider the recurrence for the minimum cost TSP tour: 6 / 6 points

$$\text{minCostTSPTour}(C) = \min \begin{cases} \text{minCostTour}(\{2, 3, 5\}, 4) + ??_1 \\ \text{minCostTour}(\{2, 3, 4\}, 5) + ??_2 \\ \text{minCostTour}(\{3, 4, 5\}, ??_3) + ??_4 \\ \text{minCostTour}(??_5, 3) + ??_6 \end{cases}$$

Select appropriate choices for the missing portions.

- ☐ $??_2 = C_{5,2}$
- ☒ $??_6 = C_{3,1}$
- ☒ Correct
- Correct.
- ☐ $??_3 = 1$
- ☒ $??_1 = C_{4,1} = 2$
- ☒ Correct
- Correct
- ☐ $??_5 = \{3, 4, 5\}$
- ☒ $??_4 = C_{2,1}$
- ☒ Correct
- Correct

